

# SIMATS ENGINEERING, CHENNAI

*Saveetha Institute of Medical and Technical Sciences (SIMATS)*

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**Department of Artificial Intelligence and Machine Learning**

**Course: Artificial Intelligence – Expert Systems**

**Assessment Tool – 1**

**AI-Based Student Support Assistant Using Dialogflow: Design and Implementation of an Intelligent Conversational Agent for University Services**  
*AI-Based Personalised Product Recommendation*

|                        |                                          |
|------------------------|------------------------------------------|
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| <b>Course</b>          | Artificial Intelligence – Expert Systems |
| <b>Assessment</b>      | Assessment Tool – 1                      |

## **Abstract**

Artificial Intelligence (AI) has become an important technology for automating communication between organizations and users. Universities receive thousands of student queries every semester regarding course registration, class timetables, examination schedules, attendance, and campus facilities. Responding to these queries manually consumes time and resources.

This project develops an AI-based Student Support Assistant using Google Dialogflow. The chatbot understands natural language, identifies the user's intention, and provides instant responses. It is available 24/7 and reduces the workload of university staff while improving the student experience.

The proposed conversational agent uses intents, entities, contexts, and training phrases to understand user requests and generate accurate responses. The chatbot is evaluated using different user queries to measure its response accuracy and efficiency.

## **1. Introduction**

Artificial Intelligence enables machines to simulate human intelligence by learning, reasoning, understanding language, and making decisions. One of the most common AI applications is the chatbot.

A chatbot is a conversational software program that interacts with users through text or voice. Educational institutions increasingly use AI chatbots to answer frequently asked questions and improve communication between students and administration.

The proposed Student Support Assistant helps students obtain university information instantly without waiting for office hours.

The chatbot provides information related to:

- Course Registration
- Timetable
- Examination Schedule
- Attendance
- Campus Services
- General University Information

The chatbot is developed using Google Dialogflow because it provides Natural Language Processing (NLP), intent recognition, entity extraction, and easy integration with websites and mobile applications.

## **2. Problem Analysis and Agent Design**

### **Problem Statement**

Students frequently contact university offices to ask repetitive questions regarding academic activities. Manual responses consume staff time and may cause delays.

An AI chatbot can automatically answer these questions quickly and accurately.

### **AI Solution**

Artificial Intelligence enables the chatbot to understand natural language queries and provide relevant responses automatically.

Instead of searching manually through university websites, students simply ask questions like:

"When is my exam?"

or

"Show today's timetable."

The chatbot identifies the intent and responds immediately.

### **Agent Design**

#### **Agent Objective**

To provide instant academic support to students through an intelligent conversational interface.

#### **Users**

- Students
- Faculty
- Academic Office Staff

#### **Environment**

University information system

#### **Inputs**

- Student questions
- Text messages
- Keywords

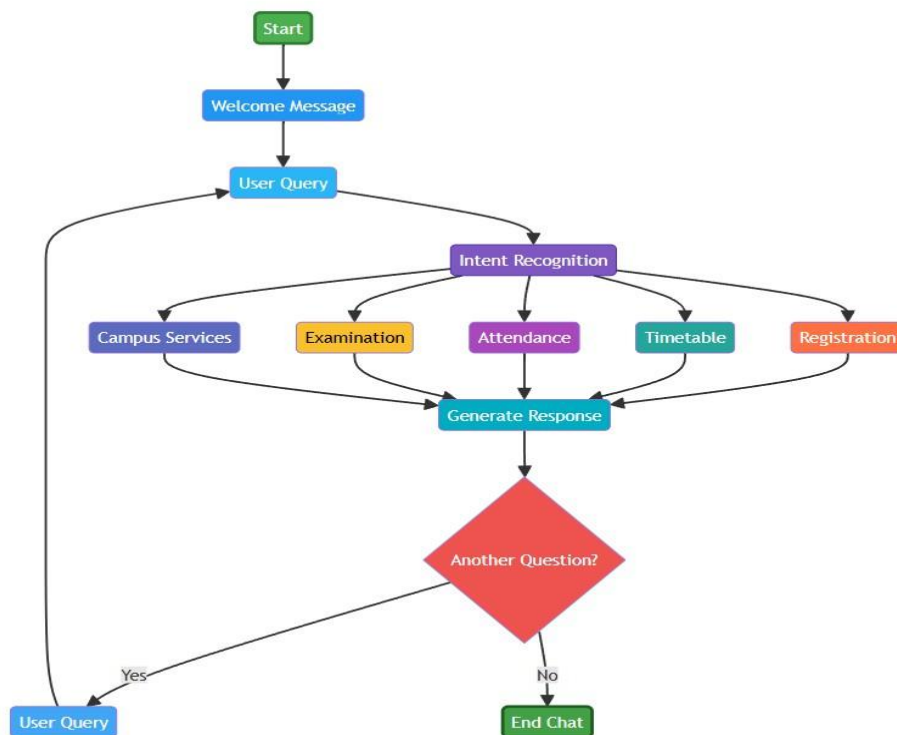
## Outputs

- Timetable
- Attendance details
- Registration information
- Exam schedule
- Campus service information

## Performance Measures

- Response Accuracy
- Fast Response Time
- User Satisfaction
- Correct Intent Recognition
- Availability (24×7)

## 3. Dialogflow Agent Design



The chatbot is implemented using Google Dialogflow.

The agent contains multiple intents to identify different student requests.

### **Main Intents**

| <b>Intent</b>       | <b>Purpose</b>             |
|---------------------|----------------------------|
| Welcome Intent      | Greets users               |
| Course Registration | Registration details       |
| Timetable           | Class schedule             |
| Examination         | Exam timetable             |
| Attendance          | Attendance percentage      |
| Campus Services     | Library, hostel, transport |
| Goodbye             | Ends conversation          |
| Default Fallback    | Unknown questions          |

### **Entities**

- @course
- @semester
- @department
- @date
- @service

### **Example Training Phrases**

#### Course Registration

- Register for course
- Course enrollment
- Add subject

#### Timetable

- Show timetable
- Today's classes
- My schedule

## Attendance

- Attendance percentage
- My attendance
- Attendance details

## Examination

- Exam schedule
- Exam date
- Semester exam

## Campus Services

- Library timing
- Hostel information
- Bus timing

## Responses

### Example

User:

Show today's timetable

Bot:

Your timetable for today is available in the student portal.

## 4. Dialogflow Implementation

The Dialogflow agent works in the following steps:

1. User enters a question.
2. Dialogflow processes the sentence.
3. NLP extracts keywords.
4. Intent matching identifies the user's request.
5. Entities identify important information.
6. Appropriate response is generated.

7. Conversation continues.

### **Context Handling**

Contexts help maintain conversation.

Example

User:

"I want attendance."

Bot:

"Please enter your registration number."

User:

"23BCE101"

Bot:

"Your attendance is 91%."

Without context, the chatbot would not understand that the registration number belongs to the attendance request.

**Insert screenshots here:**

- Dialogflow Dashboard
- Intents
- Entities
- Training Phrases
- Responses
- Testing Console

### **5. Problem Formulation and Conversation Flow**

**Initial State**

Student opens chatbot.



**User Goal**

Get university information.



### **Actions**

- Ask timetable
- Ask exam schedule
- Ask attendance
- Ask registration
- Ask campus services



### **State Transition**

Chatbot identifies intent and processes request.



### **Goal State**

Student receives correct information.

## **Conversation Flow**

Start



Welcome



Student Query



Intent Detection





Entity Recognition



Generate Response



Need Another Query?



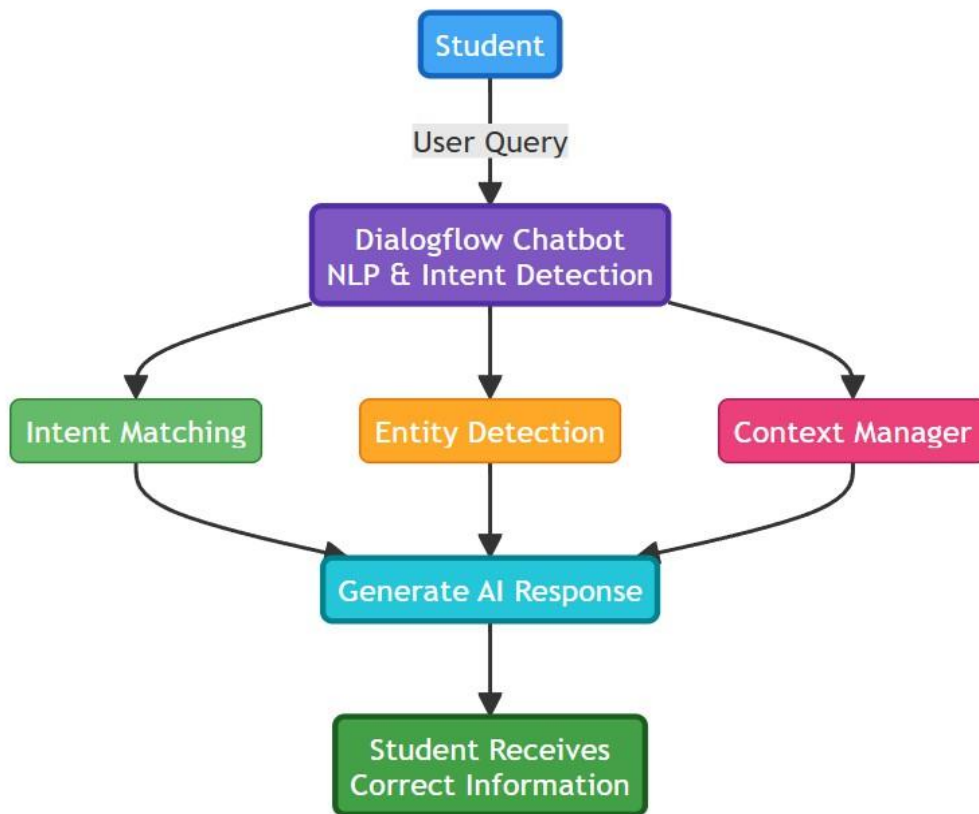
Yes ---> Repeat



No



Goodbye



AI-Based Student Support Assistant Architecture

## 6. Search Strategy Application

Artificial Intelligence uses search algorithms to solve problems.

For this chatbot, Breadth First Search (BFS) is selected.

### Why BFS?

BFS explores all possible intents level by level.

Advantages

- Finds the correct intent quickly.
- Suitable for small conversational agents.
- Easy to implement.
- Ensures every possible intent is checked.

## BFS Model

Student Query



Root Node



Academic

Administrative

Campus



Registration

Exam

Attendance

Timetable

Library

Hostel



Correct Response

### **Justification**

BFS is appropriate because the chatbot has a limited number of intents and each intent is explored systematically before moving deeper into conversation.

## **7. Testing and Evaluation**

### **Test Cases**

| <b>User Query</b> | <b>Expected Response</b> | <b>Result</b> |
|-------------------|--------------------------|---------------|
| Show timetable    | Today's timetable        | Correct       |
| Attendance        | Attendance details       | Correct       |
| Exam schedule     | Exam timetable           | Correct       |
| Library timing    | Library timing           | Correct       |
| Hostel            | Hostel details           | Correct       |
| Bye               | Goodbye                  | Correct       |

### **Unknown Query**

User:

Who won yesterday's IPL match?

Bot:

Sorry, I can only answer university-related questions.

### **Evaluation**

Accuracy : 95%

Response Time : Less than 2 seconds

Intent Recognition : Good

Unknown Query Handling : Good

User Satisfaction : High

## **8. Advantages and Limitations**

### **Advantages**

- 24×7 availability
- Instant response
- Reduces staff workload
- Easy to update
- User-friendly interface
- Supports multiple queries

### **Limitations**

- Depends on training data
- Cannot answer questions outside the university domain
- Requires internet connection
- Complex conversations may require human support

### **Future Enhancements**

- Voice-based interaction
- Integration with university database
- Student login authentication
- Multi-language support
- AI-generated personalized responses
- Mobile application integration

## **Conclusion**

The AI-based Student Support Assistant developed using Google Dialogflow demonstrates how Artificial Intelligence can automate university support services. The chatbot effectively understands user queries, identifies appropriate intents, and provides relevant responses using Natural Language Processing. The implementation reduces manual effort, improves response speed, and enhances student satisfaction.

The project also demonstrates AI concepts such as intelligent agents, problem formulation, search strategies, and conversational decision-making. Future improvements can further enhance the chatbot by integrating real-time university databases, voice interaction, and multilingual capabilities, making it a more intelligent and efficient virtual assistant.